

React HandsOn_11:-

THEORY

Explain React Events

React Events are actions performed by the user or browser, such as clicking a button, typing in a textbox, submitting a form, or moving the mouse. React uses events to make web pages interactive.

Common React Events

Event	Description
onClick	Triggered when an element is clicked
onChange	Triggered when the value of an input changes
onSubmit	Triggered when a form is submitted
onMouseOver	Triggered when the mouse moves over an element
onMouseOut	Triggered when the mouse leaves an element
onKeyDown	Triggered when a key is pressed
onFocus	Triggered when an input gains focus
onBlur	Triggered when an input loses focus

Example

```
function App() {  
  function showMessage() {  
    alert("Button Clicked!");  
  }  
  return (  
    <button onClick={showMessage}>  
      Click Me  
    </button>  
  );  
}  
  
export default App;
```

Output:

When the button is clicked, an alert box displays "Button Clicked!".

Explain about event handlers

An Event Handler is a JavaScript function that is executed when an event occurs.

Syntax

```
<element eventName={functionName} />
```

Example 1: Named Function

```
function App() {  
  function greet() {  
    alert("Welcome to React");  
  }  
  return (  
    <button onClick={greet}>  
      Click Here  
    </button>  
  );  
}
```

Example 2: Arrow Function

```
function App() {  
  return (  
    <button onClick={() => alert("Hello")}>  
      Click  
    </button>  
  );  
}
```

Example 3: Passing Arguments

```
function App() {
```

```
function display(name) {
  alert("Hello " + name);
}

return (
  <button onClick={() => display("Arpita")}>
    Click
  </button>
);
```

- Respond to user actions.
- Execute JavaScript code.
- Update the UI.
- Handle form submissions.
- **Change component state.**

Define Synthetic event

A Synthetic Event is a wrapper around the browser's native event provided by React. It behaves like a normal JavaScript event but works consistently across all browsers.

Features

- Cross-browser compatible.
- Same API as native events.
- Improves performance.
- Normalizes browser differences.

Example

```
function App() {

  function handleClick(event) {
    console.log(event.type);
```

```
    alert("Button Clicked");
  }

  return (
    <button onClick={handleClick}>
      Click
    </button>
  );
}
```

When the button is clicked:

- event is a Synthetic Event.
- event.type returns "click".

Identify React event naming convention

React event names follow a specific naming convention:

- Use camelCase.
- Prefix the event with on.
- Pass a function (not a string).

Examples

HTML	React
onclick	onClick
onchange	onChange
onsubmit	onSubmit
onmouseover	onMouseOver
onmouseout	onMouseOut
onkeydown	onKeyDown

Correct Example

```
<button onClick={handleClick}>
  Click Me
</button>
```

HANDS-ON LAB

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Say welcome

Click on me

Currency Converter!!!

Amount: 100

Currency: 300

Submit

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Increment

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Say welcome

Click on me

Currency Converter!!!

Amount: 100

Currency: 300

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Hello! Member!

OK

